

Ariun-Erdene Tumurchuluun, Curriculum Vitae

CONTACT

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RESEARCH PROFILE

M.Sc. graduate (Erasmus Mundus Joint Master in Language and Communication Technologies) and AI engineer studying how multilingual large language models represent alignment-relevant and fairness-related concepts, and how mechanistic interpretability methods can localize and control those behaviors across languages and architectures.

Published peer-reviewed workshop research on tense localization and control in a multilingual LLM (MRL @ EMNLP 2025, Best Paper Award) and currently co-author on a manuscript under review at CoLM 2026 on cross-lingual stereotype bias. Seeking a PhD to deepen work on interpretability, multilingual NLP, and trustworthy LLMs.

RESEARCH INTERESTS

Mechanistic interpretability; multilingual LLMs; representation analysis; sparse autoencoders and monosemantic features; causal tracing and probing; model steering; AI fairness, alignment, and safety.

EDUCATION

- Oct 2023 – Sep 2025 **M.Sc., Language Science and Technology**, *Universität des Saarlandes*, Saarbrücken, Germany.
Thesis: *Causal Investigation of Tense Encoding in a Multilingual Transformer*.
GPA: 2.0/5.0 (German scale; 1.0 = best).
Coursework: Multimodal Language Understanding, Mechanistic Interpretability, Speech Science, Software Project, Computational Linguistics. LCT Erasmus Mundus Joint Master's Scholarship (fully funded).
- Oct 2023 – Sep 2025 **M.Sc., Language Technologies and Computational Linguistics**, *Charles University*, Prague, Czech Republic.
Thesis: *Causal Investigation of Tense Encoding in a Multilingual Transformer*.
GPA: 1.91/4.0 (Czech scale; 1.0 = best), Summa Cum Laude.
Coursework: Deep Learning, Unsupervised Machine Learning, Speech Recognition and Generation, Statistical Methods in NLP. LCT Erasmus Mundus Joint Master's Scholarship (fully funded).

Sep 2018 – May 2022 **B.Eng., Software Engineering**, *National University of Mongolia*, Ulaanbaatar, Mongolia.
GPA: 3.8/4.0 (top 1%), Summa Cum Laude.
Coursework: Web Development, Object-Oriented Programming, Data Structures, Algorithms, Databases, Software Engineering, Project Management, Agile Development, Cloud Computing, Cybersecurity.

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed

1. **Ariun-Erdene Tumurchuluun**, Yusser Al Ghussin, David Mareček, Josef Van Genabith, and Koel Dutta Chowdhury. 2025. TenseLoC: Tense Localization and Control in a Multilingual LLM. In *Proceedings of the 5th Workshop on Multilingual Representation Learning*, pages 243–264, Suzhou, China. Association for Computational Linguistics.

Manuscripts under review

1. **Ariun-Erdene Tumurchuluun***, Yusser Al Ghussin*, Pinzhen Chen, Josef Van Genabith, and Koel Dutta Chowdhury. Under review at CoLM 2026, San Francisco, USA (* equal contribution).

SELECTED RESEARCH PROJECTS

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| Nov 2025 – present | There Is No Universal Language for Bias but a Common Fix.
Investigated whether stereotype bias in state-of-the-art LLMs is shared across languages or language-specific across model families. Designed analyses spanning gender, age, sexual orientation, physical appearance, and related attributes; manuscript under review at CoLM 2026. |
| Jun 2025 – Oct 2025 | TenseLoC: tense localization and control in a multilingual LLM.
Studied internal representations of grammatical tense (morphology, syntax, semantics) in Llama 3.1 across eight languages using probing, causal tracing, sparse autoencoders, and activation steering grounded in identified tense features. Outcome: peer-reviewed publication at MRL @ EMNLP 2025 (Best Paper Award). |
| Apr 2025 | Word-order encoding in a multilingual encoder.
Implemented diagnostic classifiers to quantify mBERT sensitivity to word-order perturbations in six typologically diverse languages. |
| Mar 2022 | Myers–Briggs types from handwriting.
Collected handwriting and self-report personality data (~50 participants), preprocessed character images, and trained a supervised classifier (~61% test accuracy). |

INDUSTRY EXPERIENCE

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| Nov 2025 – present | AI Engineer , <i>Egune AI LLC</i> , Ulaanbaatar, Mongolia.
Built Mongolian NLP systems for spelling and grammar correction, POS tagging, word sense and homonym disambiguation, translation evaluation, RAG-based document retrieval, and text rewriting. Curated and generated training data; fine-tuned models with full fine-tuning, LoRA, and QLoRA (Axolotl); evaluated quality; shipped inference with Hugging Face Transformers and vLLM; documented APIs and demos; used Playwright for targeted data collection. |
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Jun 2021 – Jul 2023 **Software Engineer**, *AI Lab LLC*, Ulaanbaatar, Mongolia.

Full-stack engineer for a financial technology client: delivered and maintained three production web applications; implemented microservice features, search improvements, and OAuth SSO (Firebase Authentication, Google Cloud Functions); migrated identity from Keycloak to Firebase Auth; maintained a Django crowdfunding platform. Stack: Python, PostgreSQL, AWS, Firebase, Redis, HTML/CSS/JavaScript, automated testing.

FUNDING AND SCHOLARSHIPS

2023 – 2025 **Erasmus Mundus Scholarship (LCT)**.

Joint master's scholarship; total support approximately EUR 55,000 (stipend, tuition, and related allowances).

AWARDS AND HONORS

- Best Paper Award, 5th Workshop on Multilingual Representation Learning @ EMNLP, 2025
- Most Advanced Graduate Award, National University of Mongolia, 2022

TECHNICAL SKILLS

ML & NLP	Python, PyTorch, Hugging Face Transformers, fine-tuning (full, LoRA, QLoRA), evaluation and error analysis, data curation, RAG pipelines, mechanistic tooling (e.g., probing, causal tracing, sparse autoencoders, steering).
Engineering	Git, Linux, unit and integration testing, REST APIs, PostgreSQL, Redis, Django, Vue.js, cloud backends (AWS, Firebase, Google Cloud Functions), Playwright.
Languages	English (professional); Mongolian (native); Japanese (beginner).
References	Available upon request.